Mutli-Satellite Scheduling & GPS Coordination

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Problem

Four satellites are orbiting Earth for a mission. The satellites launched for this mission consist of one large satellite, two medium-sized satellites, and a CubeSat. Each of the satellites has different parameters and specifications, such as energy consumption, maneuverability, storage capacity, orbital visibility, and cost, etc. The mission assigned to these satellites is to image a set of 10 high-priority landmarks on the Earth.

The four satellites have different hardware limitations. The limitations provided to us are: onboard storage capacity and visibility windows (see Tables 1 & 2). To avoid redundant data collection and to minimize storage waste, each target will be imaged by at most one satellite except for the 10th landmark, which can be imaged by all four satellites. Additionally, every satellite must be utilized in the mission.

The 10th landmark is unique as it is the only target visible to all four satellites simultaneously. This visibility allows for the application of the trilateration method, enabling precise GPS Geolocation. In this configuration, the four satellites synchronize and cross-reference data, functioning as a single system. Due to this, geolocation either succeeds or fails. Every landmark is given a weight (priority) from the range 0 to 100, scaling based on importance to the mission (see Table 3).

The goal of this mission is to find the most efficient task assignment for the satellites. We aim to maximize the "mission value" (the sum of priorities) without violating any hardware constraints or physical orbital rules.

Table 1: Target Visibility from Satellite

Visibility	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**	**T7**	**T8**	**T9**	**T10**
V1	1	0	1	0	1	0	1	0	1	1
V2	0	1	0	1	0	0	1	1	0	1
V3	1	1	0	1	0	1	0	0	1	1
V4	1	0	1	0	1	0	0	0	1	1

Table 2: Maximum Storage Capacity per Satellite

Satellite	**Max GB**
S1	2500
S2	1600
S3	1200
S4	512

Table 3: Priority and Data Size Metrics of Landmarks (Targets)

Landmarks	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**	**T7**	**T8**	**T9**	**T10**
Priority (P)	95	80	60	65	90	40	50	55	85	100
Data Size (D)	80	150	400	360	50	200	650	520	90	120

Solution

The problem is a Mixed-Integer Linear Program that is solvable through Simplex Method. From the given tables we can see that the satellites are denoted by S_i, where $i \in \{1, 2, 3, 4\}$, and the landmarks (targets) are represented as T_j, where $j \in \{1, 2, \ldots, 10\}$.
Decision variables will be denoted by $x_{i,j}$ and x_{gps}.
The visibility of the target by the satellite is denoted by $V_{i,j}$.
This problem is a maximization problem. Objective function shall be denoted by Z, representing the "maximum mission value" attainable. Z measures the highest possible sum of priorities (task completion based on relevance to the mission) you can achieve without running out of storage or visibility.

1. Decision Variables

The decision variables in this problem are the tasks given to the satellite. In this case the task of the satellites is imaging. Due to this the value of the decision variable will be a binary. When $x = 0$, the satellite is not able to image the target, meanwhile when $x = 1$, the satellite is able to image the target. There are a total of **41 decision variables** calculated by: $\sum_{i=1}^{4} \sum_{j=1}^{10} x_{i,j} + x_{\text{GPS}}$.
The GPS formation creates a singular system, hence the x_{gps} which is also a binary variable. It only occurs when all 4 satellites are viewing the target simultaneouly. $x_{gps} = 1$ means that all 4 satellites are viewing the target simultaneously, allowing them to act as one system. When $x_{gps} = 0$, at least one satellite was not about to view the target.
(note: this has nothing to do with the visibilty matrix - Table 1)
$\sum_{i=1}^{4} \sum_{j=1}^{10} x_{i,j}$ represents the (4 satellites)*(10 targets).

2. Objective Function

The Objective Function $(Z) = \sum_{i=1}^{4} \sum_{j=1}^{9} P_j * x_{i,j} + P_{10} * x_{gps}$
Here $P_{10} = 100$ (the highest priority). The satellites viewing target 10 are considered to be a single system as: $x_{gps} = \frac{1}{4} \sum_{i=1}^{4} x_{i,10} = \frac{1}{4}(x_{1,10} + x_{2,10} + x_{3,10} + x_{4,10})$
Expanded form of Objective Function:
$Z = 95 \cdot x_{1,1} + 80 \cdot x_{1,2} + 60 \cdot x_{1,3} + 65 \cdot x_{1,4} + 90 \cdot x_{1,5} + 40 \cdot x_{1,6} + 50 \cdot x_{1,7} + 55 \cdot x_{1,8} + 85 \cdot x_{1,9}) + ... + 85 \cdot x_{4,9} + 100 \cdot x_{gps}$

3. Constraints

From the problem, the total number of constraints is 59, which can be classified into 6 different types of constraints.

1. Target Allocation Constraint: Targets T_1 to T_9 are imaged at most by one satellite, this ensures that storage isn't being wasted and prevents data redundancy. T_10, the target where GPS coordination occurs can be imaged by up to four satellites to construct a 3D image.
$\sum_{i=1}^{4} X_{i,j} \leq 1, \quad \forall j \in \{1, \ldots, 9\}$

$\sum_{i=1}^{4} X_{i,10} \leq 4$

2. Storage Capacity Constraint: The total data volume collected by each satellite is limited to its maximum onboard storage capacity. Summation of D_j for the assigned targets should no exceed the hardware storage limit.
$\sum_{j=1}^{10} D_j \cdot X_{i,j} \leq CAP_i, \quad \forall i \in \{1, \ldots, 4\}$
Expanded Form:
$80 \cdot x_{1,1} + 150 \cdot x_{1,2} + 400 \cdot x_{1,3} + 360 \cdot x_{1,4} + 50 \cdot x_{1,5} + 200 \cdot x_{1,6} + 650 \cdot x_{1,7} + 520 \cdot x_{1,8} +$
$90 \cdot x_{1,9} + 120 \cdot x_{1,10} \leq 2500$
likewise...
$80 \cdot x_{2,1} + ... + 120 \cdot x_{2,10} \leq 1600$
$80 \cdot x_{3,1} + ... + 120 \cdot x_{3,10} \leq 1200$
$80 \cdot x_{4,1} + ... + 120 \cdot x_{4,10} \leq 512$

3. Visibility Constraint: A satellite can only be assigned to image a landmark if that target is within its current line-of-sight visibility window. Table 1 is used to determine that. When $V_{i,j} = 0$, that means that the target is not within the visible range of the satellite, while $V_{i,j} = 1$ means that the target lies in the visible range.
$x_{i,j} \leq V_{i,j}, \quad \forall i, j \in \{1, \ldots, 4\}$

4. No Satellite can stay Idle: Every satellite needs to be occupied with atleast one imaging task. This ensures maximum utilization of all four satellites.
$\sum_{j=1}^{10} X_{i,j} \geq 1, \quad \forall i \in \{1, \ldots, 4\}$

5. Binary Constraint: All the assignment variables, $x_{i,j}$, and x_{GPS} are binary (all decision variables are binary). This constraint needs to be added into the LINGO model and the Excel Solver Parameters. Without it we would get some value for $x_{i,j}$ which would not make sense.
$x_{i,j} \in \{0, 1\} \quad and \quad x_{GPS} \in \{0, 1\}$

6. GPS Geolocation: Every action taken by the four satellites to T_{10} is considered as a singular task as it is a single system.
$4 \cdot x_{GPS} \leq \sum_{i=1}^{4} x_{i,10}$

4. Excel Sheet Data and Results

The below image are various tables made to solve the problem using Simplex LP Solver in Excel.
The yellow cells are for decision variables and the objective function.
Utilization refers to the no-idle constraint we added.
Optimized Imaging is the constraint we added for the at most one satellite for every target case, it needs to be less than or equal to the **Limit for Imaging**.
Used GB and **Max GB** describe the Storage Capacity Constraint. XGPS and GPS refer to the GPS Geolocation constraint where the formula is mimicked as; XGPS $\leq$ GPS, where (GPS = SUM(4, T10 column)/4).

Due to the binary constraints, the Solver does not give a sensitivity report for this model due to the discrete nature of binary variables. To attain a sensitivity report, the binary constraint needs to be removed from the Solver parameters.
Overall from the solver we get our Max Z = 720.

Decision Variables											
	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Utlisation
S1	1	0	1	0	0	0	1	0	1	1	5
S2	0	1	0	1	0	0	0	1	0	1	4
S3	0	0	0	0	0	1	0	0	0	1	2
S4	0	0	0	0	1	0	0	0	0	1	2
Optimised Imaging	1	1	1	1	1	1	1	1	1	4	-
Limit for Imaging	1	1	1	1	1	1	1	1	1	4	-

Used GB	Max GB
1340	2500
1150	1600
320	1200
170	512

Priority (P)	95	80	60	65	90	40	50	55	85	100
Data Size (D)	80	150	400	360	50	200	650	520	90	120

Max Z	720

Visibility										
VS1	1	0	1	0	1	0	1	0	1	1
VS2	0	1	0	1	0	0	1	1	0	1
VS3	1	1	0	1	0	1	0	0	1	1
VS4	1	0	1	0	1	0	0	0	1	1

X_GPS	1
GPS	1

Figure 1: Data from Excel

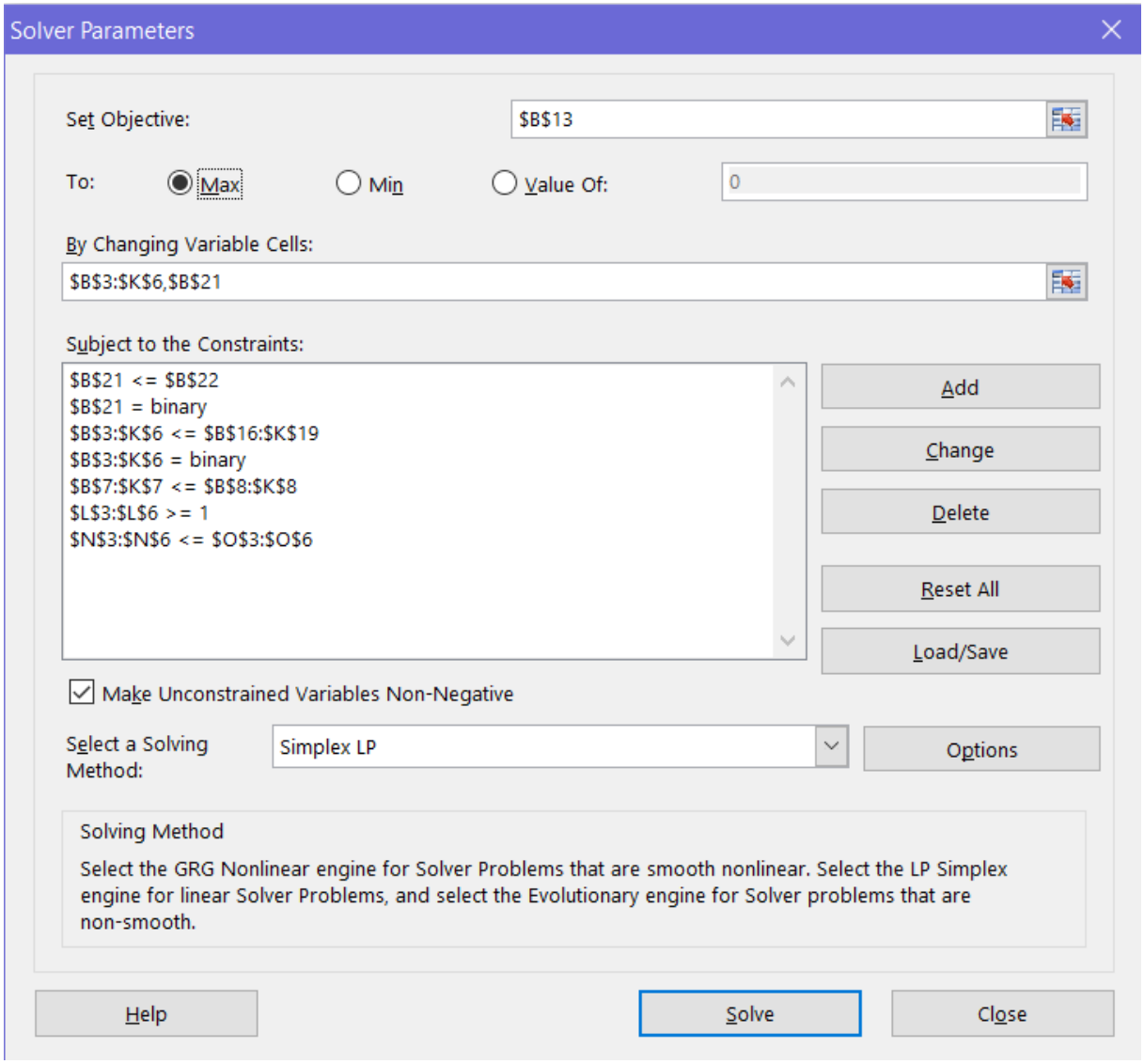


Figure 2: Solver Parameters from Excel

5. LINGO Code

The code below is the model implemented in LINGO to solve this linear progression problem on multi-satellite scheduling.

```
MODEL:
! Multi-Satellite Scheduling and GPS Coordination;

SETS:
  SATS /1..4/: CAP;              ! Storage Capacity;
  TARGETS /1..10/: P, D;         ! Priority and Data Size;
  LINKS(SATS, TARGETS): V, X; ! Visibility and Binary Variables;
ENDSETS

DATA:
  CAP = 2500, 1600, 1200, 512;
  P = 95, 80, 60, 65, 90, 40, 50, 55, 85, 100;
  D = 80, 150, 400, 360, 50, 200, 650, 520, 90, 120;
  V = 1,0,1,0,1,0,1,0,1,1
      0,1,0,1,0,0,1,1,0,1
      1,1,0,1,0,1,0,0,1,1
      1,0,1,0,1,0,0,0,1,1;

ENDDATA

! OBJECTIVE FUNCTION;
MAX = @SUM(TARGETS(J) | J #LT# 10: P(J) * @SUM(SATS(I): X(I, J)))
    + P(10)*X_GPS;


! CONSTRAINTS;

! 1. At most one Satellite per Target;
@FOR(TARGETS(J) | J #LT# 10:
   @SUM(SATS(I): X(I, J)) <= 1
);

! 2. Maximum 4 Satellites for Target 10;
@SUM(SATS(I): X(I,10)) <= 4;

! 3. Maximum Satellite Storage Limits;
@FOR(SATS(I):
   @SUM(TARGETS(J): D(J) * X(I, J)) <= CAP(I)
);

! 4. Visibility Constraints;
@FOR(LINKS(I, J):
   X(I, J) <= V(I, J)
);

! 5. No Satellite can be idle;
@FOR(SATS(I):
@SUM(TARGETS(J): X(I, J)) >= 1
);

! 6. Decision Variables = Binary;
```

```
@FOR(LINKS(I, J): @BIN(X(I, J)));
@BIN(X_GPS);

! GPS coordination - Single System formation;
4*X_GPS <= @SUM(SATS(I): X(I, 10));

END
```

Listing 1: LINGO Model

After pressing solve, the solver solves the thing and we get $Z = 720$, just like we got in the excel file.

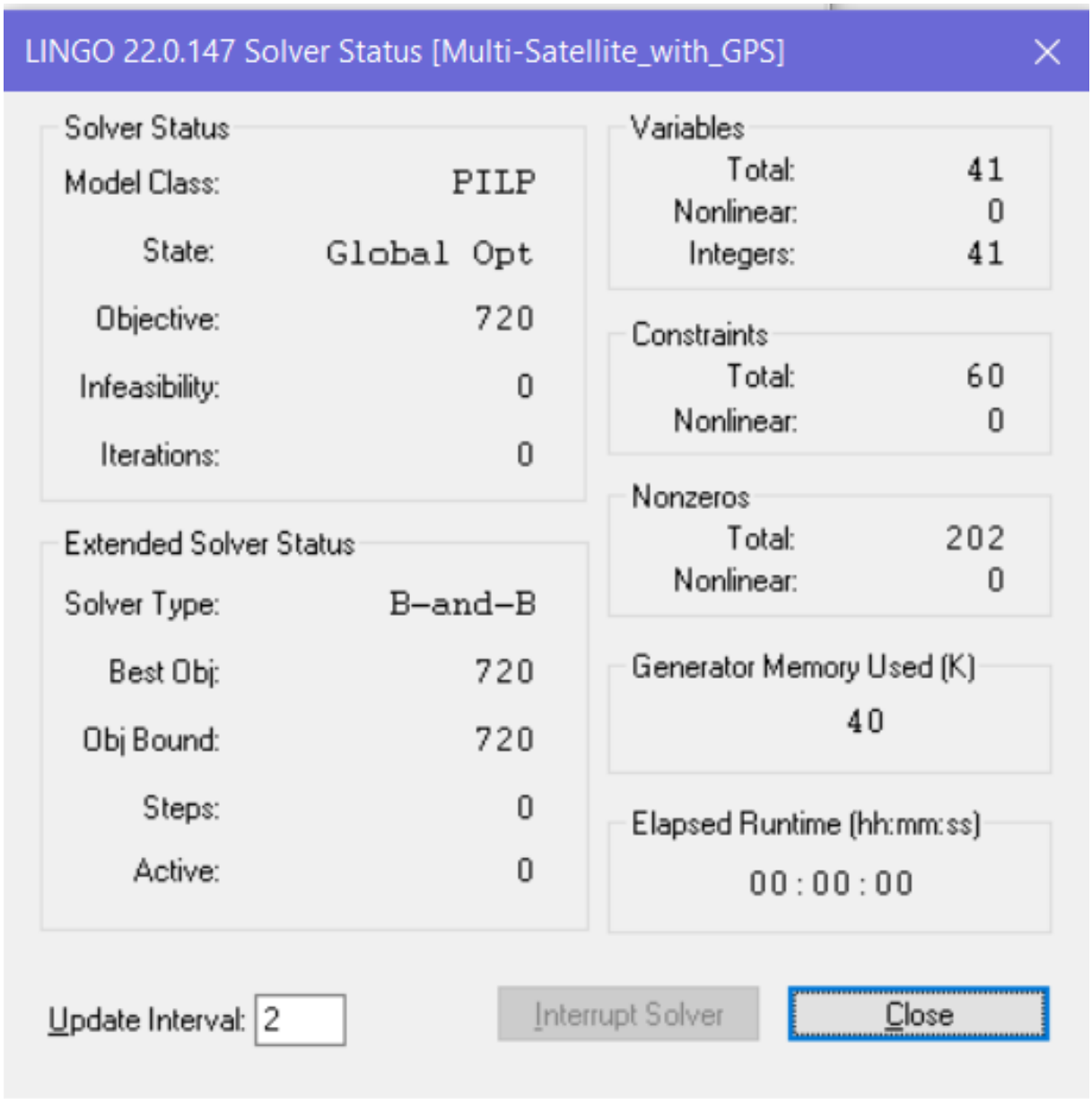


Figure 3: Solution from LINGO

Conclusion

In this study, a multi-satellite scheduling problem was successfully modeled and solved using Mixed-Integer Linear Programming (Simplex Method) in both Excel and LINGO. The goal to maximize the mission value through the increase in priority was achieved. GPS Geolocation was also implemented and used. The results demonstrate that the setup

was able to reach an optimal value of 720. The mission value could have increased if instead of a CubeSat (mini satellite), a larger satellite of >2000GB could've been launched instead. Satellite 2 with the 1600 GB could've either changed it's orbit trajectory or a different satellite could've been sent so that landmarks 1 and 9 are visible. If these two cases are achieved then the mission value could increase as the data sizes for those targets are comparatively less. Another reason is that larger satellites can perform more imaging tasks due to their larger onboard storage.

(note: Sensitivity Report is attached in the Excel file (binary constraints are removed for that one case)

Table 4: Summary of Results

Metric	**Value**
Objective Value (Z)	720
Total Constraints	60
Total Variables	41
Integer Variables	41
Non-zeroes	202